The Physician and Sportsmedicine

ISSN: (Print) (Online) Journal homepage: https://www.tandfonline.com/loi/ipsm20

Epidemiology of Youth and High School American Football-Related Injuries Presenting to United States Emergency Departments: 2010-2019

Connor Hoge, Ramsey Sabbagh, Malik Morgan & Brian M. Grawe

To cite this article: Connor Hoge, Ramsey Sabbagh, Malik Morgan & Brian M. Grawe (2021): Epidemiology of Youth and High School American Football-Related Injuries Presenting to United States Emergency Departments: 2010-2019, The Physician and Sportsmedicine, DOI: 10.1080/00913847.2021.1931980

To link to this article: https://doi.org/10.1080/00913847.2021.1931980

Accepted author version posted online: 24 May 2021.

Submit your article to this journal

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=ipsm20

Check for updates

Publisher: Taylor & Francis & Informa UK Limited, trading as Taylor & Francis Group

Journal: *The Physician and Sportsmedicine*

DOI: 10.1080/00913847.2021.1931980

Epidemiology of Youth and High School American Football-Related Injuries Presenting to United States Emergency Departments: 2010-2019

Epidemiology of Youth and High School American Football-Related Injuries Presenting to United States Emergency Departments: 2010-2019

Running Title: Youth and High School American Football Injuries

Connor Hoge, BS[1]; Ramsey Sabbagh, BS[2]; Malik Morgan, BS[3]; Brian M. Grawe, MD[4]

[1]Connor Guernica Hoge, BS
University of Cincinnati
Department of Orthopaedics and Sports Medicine
231 Albert Sabin Way
Cincinnati, OH 45529

[2]Ramsey Samir. Sabbagh, BS
University of Cincinnati
Department of Orthopaedics and Sports Medicine
231 Albert Sabin Way
Cincinnati, OH 45529

[3]Malik Morgan, BS
University of Cincinnati
Department of Orthopaedics and Sports Medicine
231 Albert Sabin Way
Cincinnati, OH 45529

[4]Brian M. Grawe, MD
University of Cincinnati
Department of Orthopaedics and Sports Medicine
231 Albert Sabin Way
Cincinnati, OH 45529

ACCEPTED MANUSCRIPT

Corresponding Author:
Connor Hoge, BS
University of Cincinnati
Department of Orthopaedics and Sports Medicine
231 Albert Sabin Way
Cincinnati, OH 45529
Email: hogecg@mail.uc.edu
Mobile: (614) 578-6164

ACCEPTED MANUSCRIPT

ABSTRACT

Objectives: American football is one of the most popular sports in the United States (US) among youth and high school athletes; however, participation is decreasing due to concerns about safety. This has led to many new safety initiatives over the last decade, but few studies have sought to evaluate if these interventions have made football safer. The purpose of this study is to investigate national-level data on football-related injuries to compare the injury characteristics and specific injury risks of high school and youth athletes over the last decade.

Methods: This cross-sectional, retrospective study queried the National Electronic Injury Surveillance System (NEISS) database from 2010-2019 to compare national weighted estimates and injury characteristics of youth (ages 6-12) and high school (ages 14-18) athletes presenting to US emergency departments (EDs) with football-associated injuries.

Results: The overall injury risk of football-related injuries presenting to US EDs for high school athletes was 16.17 (C.I. 13.85-18.38) per 1,000 participants/year compared to 11.97 (C.I. 10.34-13.41) per 1,000 participants/year in youth athletes. From 2010 to 2019, the overall injury risk for youth athletes participating in football significantly ($p < 0.0001$) decreased from 13.31 (C.I. 11.23-15.38) per 1,000 participants to 9.93 (C.I. 7.24-12.62) per 1,000 participants. Additionally, the overall injury risk for high school athletes participating in football significantly ($p < 0.0001$) decreased from 2010 (N = 18.63 [C.I. 16.05-21.21] per 1,000 participants) to 2018 (N = 13.19 [C.I. 9.94-16.44] per 1,000 participants). High school athletes were almost twice as likely and 2.37 times more likely to present with knee and shoulder injuries compared to youth athletes per 1,000 participants/year.

ACCEPTED MANUSCRIPT

Conclusion: Our study provides evidence that the overall injury risk for both youth and high school football players has significantly decreased since 2010. Furthermore, our study demonstrates that these athletes sustain unique injury characteristics.

KEYWORDS: Epidemiology; Football; Youth; High School; Sports

ACCEPTED MANUSCRIPT

Introduction

American football is one of the most popular sports in the United States (US) among youth and high school athletes. Over the last decade, however, there has been a steady decline in participation numbers for both of these groups. According to the Sports and Fitness Industry Association (SFIA), a trade organization that has established itself as the gold standard in youth participation research-objective data in conjunction with the Project Play report, the participation numbers for youth athletes between the ages of 6 and 12 has dropped an astounding 18.7% since 2010 [1]. The National Federation of State High School Associations (NFHS), which has been collecting annual sport participation data in high school athletes since 1969, has noted a 9.1% decrease in high school football participation since 2010 [2].

Much of this decline in participation coincides with public debate over the safety of the sport and recent medical research showing the long-term harm that can be caused by concussions [3,4]. A number of states, including California, Illinois, New York and Maryland, have introduced legislation proposing a minimum of 12 years of age for tackle football participation [5]. Additionally, both national and state-level organizations, such as Pop Warner, have begun to implement rule changes to make tackle football safer. Irrespective of these efforts, a recent survey in 2017 found that 78% of US adults do not think children under the age of 14 should play tackle football [6].

Given the changes in participation and the shift towards increased injury prevention measures over the last decade, there is a paucity of recent studies describing injury patterns and injury rates in youth and high school athletes [7,8]. One study between 2005-2014 demonstrated a relatively stable injury rate in high school football, around 13 injuries per 1,000 athletic

ACCEPTED MANUSCRIPT

exposures (AEs) [7]. Many of the safety initiatives that are commonplace in football today were implemented after 2010, so it is possible that there was not enough time elapsed in this study to demonstrate the true impact of these safety precautions [5]. With a majority of recent data looking at the impact of youth and adolescent head injuries, there remains little data representing overall injury trends and injury characteristics in youth football. The purpose of this study is to investigate national-level data on football-related injuries to evaluate the injury characteristics between high school and youth athletes. Furthermore, we hope to use recent participation data from these groups to compare injury rates and analyze overall injury trends to further aid in appropriate injury prevention practices.

ACCEPTED MANUSCRIPT

Materials and Methods

This cross-sectional, retrospective study identified cases of football-related injuries in youth and high school age patients using the Consumer Product Safety Commission's (CPSC) National Electronic Injury Surveillance System (NEISS). The database is publicly available, deidentified, and published annually on a freely accessible governmental website. Additionally, it is a nationally representative probability sample of designated hospital emergency departments (EDs) stratified by hospital size and geographic location, from which weighted national estimates and sampling errors for queried injuries may be derived. A wide variety of reliable, reproducible epidemiologic studies on injury-related emergency department visits have been published using the NEISS database [9,10].

Selection Criteria

In this investigation, each yearly sample in the NEISS database was queried between 2010 and 2019, both inclusive, for injuries classified as associated with the sport of football (Product Code: 1211 – "Football (activity, Apparel Or Equipment)"). A total of 102,944 unique cases were identified in the NEISS database during this period.

Next, cases in which patients 13 years of age were excluded from our analysis. The decision to exclude this age group was two-fold. First, participation data for youth and high school football is not readily available. The most accurate and reliable data for youth participation is provided by the Sports & Fitness Industry Association (SFIA) [1]. In their annual reports, they define youth as ages 6 to 12. Furthermore, the most accurate high school participation numbers are provided by the National Federation of State High School Associations (NFHS) [2]. The typical high school athlete participating is between 14 and 18 years of age.

ACCEPTED MANUSCRIPT

Therefore, the decision to define our age groups as 6 to 12 (youth) and 14 to 18 (high school) allows for more accurate use and interpretation of participation numbers. Lastly, excluding patients 13 years of age helps to more clearly delineate between youth and high school participants as it reduces the chance of overlap between the two groups. After limiting our analysis to those between 6 and 12 years of age and 14 and 18 years of age, a total of 89,236 unique cases were identified in the NEISS database during this period, which amounted to 2,827,751 weighted national estimates of football-related injuries presenting to US EDs.

Data Management

In order to facilitate analysis, some of the smaller body part and diagnosis categories constituting less than 1% of the total injury count were consolidated into larger categories. 'Foot and Ankle' included foot, ankle, and toe injuries, while 'Hand and Wrist' included wrist, hand, and finger injuries, and 'Head and Neck' included head, neck, face, ear, eyeball, and mouth injuries. 'All Other Diagnoses' included nerve damage, amputation, aspiration, burns, crushing, dental, dermal/conjunctival, foreign body, hematoma, hemorrhage, ingestion, poisoning, puncture, radiation, submersion, and other injuries. The top five most commonly injured body parts in youth and high School tackle football were selected for Poisson regression analysis due to the formidable burden of these injuries within this population. All data was queried, extracted, and managed by an experienced statistician (RS).

Statistical Analyses

Weighted national estimates and 95% confidence intervals (C.I.) were calculated using IBM SPSS Statistics for Windows, version 27.0 [11]. Bivariate comparisons between variables were conducted using chi-square analysis with strength of association assessed using odd ratios

ACCEPTED MANUSCRIPT

(OR) with 95% C.I. In cases where there are more than two categories, all odds ratios constitute a comparison between the group of interest and all other categories consolidated into a single group. Poisson regression models were used to evaluate whether youth and high school football participation counts, weighted national injury estimates, and annual injury risk significantly changed over the 2010 to 2019 time period. Injury risk was calculated using SFIA participation data for youth athletes and NFHS data for high school athletes. The number of football-related injuries per 1,000 participants/year was calculated using this participation data and the NEISS data between 2010-2019. We identified this outcome as the best representation of a gross estimate of injury risk given that lack of available exposure data. This was described as a reliable and valid outcome measure by the International Olympic Committee Injury and Illness Epidemiology Consensus Group in 2020 [12]. Participation data from the NFHS was not reported in 2019-2020 due to the COVID-19 pandemic. Bonferroni correction was used to account for testing 20 body part and diagnosis related outcomes; thus, these associations were considered significant for p-values ≤ 0.0025 (0.05/20). All other associations were considered significant for p-values ≤ 0.05.

ACCEPTED MANUSCRIPT

Results

Figure 1 demonstrates the annual number of injuries vs annual participation numbers in the US for youth and high school football between 2010-2019. From 2010 to 2019, the number of US participants in youth football decreased by an average of 1.83% (C.I. 1.81%-1.85%) per year ($p < 0.001$), while high school football participation decreased by an average of 1.19% (C.I. 1.16%-1.21%) per year ($p < 0.001$). The weighted national estimates of youth football-related injuries presenting to US EDs between 2010 and 2019 decreased by an average of 5.27% (C.I. 4.17%-6.89%) per year ($p < 0.001$). Additionally, the estimated number of high school football-related injuries decreased by an average of 4.92% (C.I. 3.85%-6.40%) per year ($p < 0.001$) during this same time period.

There was an estimated total of 2,827,751 (C.I. 2,331,892-3,323,609) football-related injuries among athletes ages 6-12 and 14-18 presenting to US EDs between 2010-2019. Youth athletes accounted for 40.13% (C.I. 38.56%-41.70%) of the injuries while high school athletes accounted for 59.87% (C.I. 58.30%-61.44%), respectively. Complete demographic characteristics of youth and high school football athletes presenting to US EDs between 2010-2019 are presented in **Table 1**.

Comparisons of national estimates of injuries in youth and high school football players stratified by body part and diagnosis are demonstrated in **Table 2**. Head/neck and hand/wrist injuries were the most common body parts injured in both youth (24.35% [C.I. 22.63%-26.17%]; 27.62% [C.I. 26.23%-29.05%]) and high school (23.35% [C.I. 21.79%-25.00%]; 20.67% [C.I. 19.57%-21.78%]) athletes. Youth athletes were 0.54 times and 0.72 times less likely to present to US EDs for shoulder and knee injuries compared to high school athletes, respectively ($p <$

ACCEPTED MANUSCRIPT

0.0001). Strains/sprains were more common in high school (29.96% [C.I. 27.62%-32.41%]) compared to youth athletes (25.85% [C.I. 23.64%-28.19%]), while fractures (23.22% [C.I. 21.98%-24.50%] vs 18.12% [C.I. 17.33%-18.93%]) were more common in youth participants. Youth athletes were 0.28 times less likely to sustain a dislocation compared to high school athletes ($p < 0.0001$).

Youth athletes were 2.49 times more likely to sustain a shoulder fracture compared to high school athletes ($p < 0.0001$). These injuries accounted for 43.27% (C.I. 39.28%-47.34%) of all football-related shoulder injuries presenting to US EDs in youth athletes. Youth athletes were 0.11 times less likely to sustain a football-related shoulder dislocation compared to high school athletes ($p < 0.0001$), which accounted for only 2.51% (C.I. 1.76%-3.57%) of football-related shoulder injuries compared to 14.97% (C.I. 12.76%-17.48%) in their high school counterparts. **Table 3** further stratifies injury diagnosis by shoulder-related injuries presenting to US EDs between 2010-2019.

Table 4 demonstrates the injury risk per 1,000 participants/year and the injury risk ratio of high school athletes compared to youth athletes presenting to US EDs with football-related injuries over the study period. High school athletes were almost twice as likely and 2.37 times more likely to present to US EDs with knee and shoulder injuries compared to youth athletes per 1,000 participants/year. There were 3.79 (C.I. 3.18-4.37) per 1,000 high school participants/year who sustained head/neck injuries necessitating ED examination over our study period. High school athletes were almost twice as likely and 5.11 times more likely to sustain a football-related concussion or dislocation compared to youth athletes per year. The overall injury risk of football-related injuries presenting to US EDs for high school athletes was 16.17 (C.I. 13.85-

ACCEPTED MANUSCRIPT

18.38) per 1,000 participants/year compared to 11.97 (C.I. 10.34-13.41) per 1,000 participants/year in youth athletes.

From 2010 to 2019, the overall injury risk for youth athletes participating in football significantly ($p < 0.0001$) decreased from 13.31 (C.I. 11.23-15.38) per 1,000 participants to 9.93 (C.I. 7.24-12.62) per 1,000 participants. Additionally, the overall injury risk for high school athletes participating in football significantly ($p < 0.0001$) decreased from 2010 (N = 18.63 [C.I. 16.05-21.21] per 1,000 participants) to 2018 (N = 13.19 [C.I. 9.94-16.44] per 1,000 participants). **Table 5** demonstrates the annual injury risk per 1,000 participants and injury risk ratio of high school athletes compared to youth athletes from 2010-2019.

ACCEPTED MANUSCRIPT

Discussion

Previous studies on youth and high school football injuries are mostly confined to examining small cohorts from local leagues or single hospitals [13-15]. Nationally representative data comparing the injury characteristics and injury rates between youth and high school football remains sparse [8,16]. A study by Mello et al. researched such injuries presenting to US EDs between 2001 and 2005 [8]. With the increased focus on injury prevention and implementation of safety initiatives over the last decade, an updated investigation of injury trends and the overall injury risk between youth and high school football participants is warranted.

Our findings demonstrate that football-related injuries presenting to US EDs pose a significant injury burden, with over 2.8 million ED visits for youth and high school aged athletes between 2010 and 2019. However, this study reveals that the number of football-related injuries presenting to US EDs significantly decreased between 2010 and 2019 by an average of 5.27% and 4.92% per year for youth and high school athletes. This coincides with a significant decrease of 1.83% and 1.19% per year in participation numbers of youth and high school athletes, respectively, most likely related to concerns about safety. Lykissas et al. found that while the injury rates of most youth sports have decreased, football injuries increased by 23% between 2000 and 2010 [17]. Since 2010, however, our study provides evidence that the overall injury risk has significantly decreased in both youth and high school football. Thus, it appears that the efforts over the last decade to make football safer, such as rule changes, improved safety equipment, and coaching proper tackling technique, are beginning to come to fruition.

Injuries to young athletes can have significant consequences later in life based on the unique physiology of growing children, including early osteoarthritis and limb length discrepancies [18,19]. Our findings demonstrate that youth and high school athletes sustain different injury characteristics, most notably high school athletes having a higher percentage of

ACCEPTED MANUSCRIPT

shoulder and elbow injuries while youth athletes sustain more hand/wrist injuries. Further investigation into shoulder injuries demonstrated that youth athletes were almost 10 times less likely to sustain a shoulder dislocation compared to high school athletes and 2.5 times more likely to sustain a shoulder fracture. In addition, we found that high school football players sustained a higher percentage of strains/sprains and a lesser percentage of fractures compared to youth football players, respectively. This is most likely due to the differences between growing and skeletally mature bone that occur within these age groups, with growing bone being more susceptible to growth plate shearing injuries and fractures due to the open physes [20].

Young athletes, when compared to their older counterparts, have a high surface-to-mass ratio, open physes, and immature motor skills, all of which can contribute to greater injury [21]. However, there is evidence that sexual maturity is associated with a higher injury risk in football [22]. Our study found that high school football players have an injury risk of 16.17 per 1,000 participants/year compared to 11.97 per 1,000 participants/year in youth football players. Most notably, high school football players were at an increased risk for shoulder and knee injuries per participants/year compared to youth football players, respectively. Furthermore, high school football players had almost double the risk of concussions and over 5 times the risk for dislocations. Previous reports found that high school football athletes have over twice as many athletic exposures per athlete-year compared to youth football players, which may predispose them to overuse injuries and an increased injury risk [23]. Additionally, there are rules in place at the youth level that are not common in high school, such as no kick-offs and weight limits for ball carriers and defenders, that are implemented to increase safety. Although football safety is trending in the right direction overall, continued attention should be focused on preventing some of the common types of injuries in these athletes.

ACCEPTED MANUSCRIPT

Many of the limitations of this study are inherent to the NEISS database. First, the true number of football-related injuries is underreported in this database because only injuries that required ED visits are recorded. Therefore, our study most likely underreports the true national burden of football-related injuries in youth and high school athletes, instead emphasizing more severe injuries. Second, the database does not code for multiple injuries in a single ED encounter. The NEISS database only codes for the most severe injury, therefore multiple injuries sustained by a single patient would not be recorded. Lastly, participation numbers provided by the SFIA and NFHS only accounted for athletes who participated in football as either a "core participant" (e.g., 26+ times/year) or on a high school team. Individuals who get injured playing football recreationally would be reported in the NEISS dataset but not accounted for in our participation numbers. Thus, our injury rates may be exaggerated due to the lack of more accurate participation numbers.

ACCEPTED MANUSCRIPT

Conclusion

Our study provides evidence that the overall injury risk for both youth and high school football players has significantly decreased since 2010. Although football carries an inherent risk of injury, the overall decrease in youth and adolescent risk of injury is reassuring. Continued attention to injury prevention, coaching, and safety precautions should be tailored for specific age groups given their unique injury characteristics and risks.

ACCEPTED MANUSCRIPT

Acknowledgements

There are no funding sources or other acknowledgements to report for the completion of this study

ACCEPTED MANUSCRIPT

Declaration of Interest

No relevant conflict of interests for any of the authors

ACCEPTED MANUSCRIPT

References

1. Aspen Institute. State of play: Trends and developments in youth sports. 2019:1-30.

2. NFHS. 2018-19 High School Athletics Participation Survey. *High Sch Particip Surv Arch*. 2019:72. https://www.nfhs.org/media/1020412/2018-19_participation_survey.pdf.

3. Marar M, McIlvain NM, Fields SK, Comstock RD. Epidemiology of concussions among united states high school athletes in 20 sports. *Am J Sports Med*. 2012. doi:10.1177/0363546511435626

4. Murphy AM, Askew KL, Sumner KE. Parents' intentions to allow youth football participation: Perceived concussion risk and the theory of planned behavior. *Sport Exerc Perform Psychol*. 2017. doi:10.1037/spy0000102

5. Analysis SP. FLAG BECOMES THE STANDARD WAY OF What if ….

6. Talty FT, Gerson J. UMass Lowell Center for Public Opinion Research Citizen Attitudes about Sports , Concussions and CTE. 2016:4-6.

7. Kerr ZY, Wilkerson GB, Caswell S V., et al. The first decade of web-based sports injury surveillance: Descriptive epidemiology of injuries in United States high school football (2005-2006 through 2013-2014) and National collegiate athletic association football (2004-2005 through 2013-2014). *J Athl Train*. 2018. doi:10.4085/1062-6050-144-17

8. Mello MJ, Myers R, Christian JB, Palmisciano L, Linakis JG. Injuries in youth football: National emergency department visits during 2001-2005 for young and adolescent players. *Acad Emerg Med*. 2009. doi:10.1111/j.1553-2712.2009.00357.x

9. Pirruccio K, Hoge C, Kelly, IV JD. Comparison of in-season and off-season wrestling injuries presenting to United States emergency departments: 2000-2018. *Phys Sportsmed*.

ACCEPTED MANUSCRIPT

2021. doi:10.1080/00913847.2020.1865780

10. Pirruccio K, Kelly JD. Weightlifting Shoulder Injuries Presenting to US. Emergency Departments: 2000-2030. *Int J Sports Med*. 2019. doi:10.1055/a-0927-6818

11. IBM Corp. IBM SPSS Statistics for Macintosh, version 27.0. *IBM Softw Bus Anal*. 2020.

12. Bahr R, Clarsen B, Derman W, et al. International Olympic Committee Consensus Statement: Methods for Recording and Reporting of Epidemiological Data on Injury and Illness in Sports 2020 (Including the STROBE Extension for Sports Injury and Illness Surveillance (STROBE-SIIS)). *Orthop J Sport Med*. 2020. doi:10.1177/2325967120902908

13. Stuart MJ, Morrey MA, Smith AM, Meis JK, Ortiguera CJ. Injuries in youth football: A prospective observational cohort analysis among players aged 9 to 13 years. *Mayo Clin Proc*. 2002. doi:10.4065/77.4.317

14. Peterson AR, Kruse AJ, Meester SM, et al. Youth football injuries: A prospective cohort. *Orthop J Sport Med*. 2017. doi:10.1177/2325967116686784

15. Smith PJ, Hollins AM, Sawyer JR, Spence DD, Outlaw S, Kelly DM. Characterization of American Football Injuries in Children and Adolescents. *J Pediatr Orthop*. 2018. doi:10.1097/BPO.0000000000001101

16. Nation AD, Nelson NG, Yard EE, Comstock RD, McKenzie LB. Football-related injuries among 6- to 17-year-olds treated in US emergency departments, 1990-2007. *Clin Pediatr (Phila)*. 2011. doi:10.1177/0009922810388511

17. Lykissas MG, Eismann EA, Parikh SN. Trends in pediatric sports-related and recreation-related injuries in the United States in the last decade. *J Pediatr Orthop*. 2013.

ACCEPTED MANUSCRIPT

doi:10.1097/BPO.0000000000000099

18. Roser LA, Clawson DK. Football injuries in the very young athlete. *Clin Orthop Relat Res*. 1970. doi:10.1097/00003086-197003000-00023

19. Marshall SW, Golightly YM. Sports injury and arthritis. *N C Med J*. 2007.

20. Caine D, Purcell L, Maffulli N. The child and adolescent athlete: A review of three potentially serious injuries. *BMC Sports Sci Med Rehabil*. 2014. doi:10.1186/2052-1847-6-22

21. Adirim TA, Cheng TL. Overview of injuries in the young athlete. *Sport Med*. 2003. doi:10.2165/00007256-200333010-00006

22. Linder MM, Townsend DJ, Jones JC, Balkcom IL, Anthony CR. Incidence of adolescent injuries in junior high school football and its relationship to sexual maturity. *Clin J Sport Med*. 1995. doi:10.1097/00042752-199507000-00006

23. Morris K, Simon JE, Grooms DR, Starkey C, Dompier TP, Kerr ZY. The epidemiology of overuse conditions in youth football and high school football players. *J Athl Train*. 2017. doi:10.4085/1062-6050-52.10.04

ACCEPTED MANUSCRIPT

Tables

Table 1: Characteristics of Youth and High School Football Players Presenting to US Emergency Departments 2010-2019		
Characteristic	**National Case Estimate**	**Percentage**
Sex		
Male	2,693,146	95.24%
Female	134,534	4.76%
Age		
Youth (6 to 12)	1,134,885	40.13%
High School (14 to 18)	1,692,866	59.87%
Race		
White	1,269,265	44.89%
Black	557,469	19.71%
Other	201,436	7.12%
Not Specified	799,580	28.28%
Disposition		
Treated and Released	2,745,829	97.10%
Treated and Admitted	39,915	1.41%

ACCEPTED MANUSCRIPT

Table 2: Comparison of National Estimates of Injuries in Youth and High School Football Stratified by Body Part and Diagnosis

	Youth		High School					
Injury	**National Case Estimate**	**%**	**National Case Estimate**	**%**	**Odds Ratio**	**Lower 95% CI**	**Upper 95% CI**	**P-value**
Body Part								
Head & Neck	276,396	24.35%	395,348	23.35%	1.06	1.05	1.06	<0.0001
Shoulder	69,754	6.15%	183,335	10.83%	0.54	0.53	0.54	<0.0001
Elbow	41,704	3.67%	42,179	2.49%	1.49	1.47	1.51	<0.0001
Lower Arm	67,895	5.98%	52,590	3.11%	1.99	1.96	2.01	<0.0001
Hand & Wrist	313,450	27.62%	349,664	20.67%	1.47	1.46	1.47	<0.0001
Upper Trunk	45,562	4.01%	81,718	4.83%	0.82	0.82	0.83	<0.0001
Lower Trunk	33,490	2.95%	71,369	4.22%	0.69	0.68	0.70	<0.0001
Upper Leg	11,415	1.01%	19,103	1.13%	0.89	0.87	0.91	<0.0001
Knee	88,000	7.75%	176,807	10.44%	0.72	0.71	0.73	<0.0001
Lower Leg	32,814	2.89%	61,795	3.65%	0.79	0.78	0.80	<0.0001
Foot & Ankle	128,387	11.31%	220,974	13.05%	0.85	0.84	0.86	<0.0001
All Other Body Parts	26,019	2.29%	37,982	2.24%	1.02	1.01	1.04	0.0067
Diagnosis								
Strain Sprain	293,355	25.85%	507,186	29.96%	0.81	0.81	0.82	<0.0001
Concussion	80,722	7.11%	157,947	9.33%	0.74	0.74	0.75	<0.0001
Contusion/Abrasion	192,939	17.00%	244,762	14.46%	1.21	1.20	1.22	<0.0001
Dislocation	14,421	1.27%	74,315	4.39%	0.28	0.28	0.29	<0.0001
Fracture	263,477	23.22%	306,669	18.12%	1.37	1.36	1.38	<0.0001
Internal Injury	93,897	8.27%	112,708	6.66%	1.27	1.25	1.28	<0.0001
Laceration	46,627	4.11%	59,727	3.53%	1.17	1.16	1.19	<0.0001
All Other Diagnoses	149,447	13.17%	229,551	13.56%	0.97	0.96	0.97	<0.0001
Total	**1,134,885**		**1,692,866**					

ACCE

Table 3: Comparison of National Estimates of Shoulder Injuries in Youth and High School Football stratified by Diagnosis

	Youth		High School					
Shoulder Diagnosis	**National Case Estimate**	**%**	**National Case Estimate**	**%**	**Odds Ratio**	**Lower 95% CI**	**Upper 95% CI**	**P-value**
Strain/Sprain	17,913	25.68%	60,251	32.86%	0.75	0.74	0.77	<0.0001
Dislocation	1,754	2.51%	27,438	14.97%	0.11	0.10	0.11	<0.0001
Contusion	12,880	18.46%	25,984	14.17%	1.10	1.08	1.13	<0.0001
Fracture	30,181	43.27%	44,006	24.00%	2.49	2.45	2.54	<0.0001

ACCEPTED MANUSCRIPT

Table 4. Injury Risk (per 1,000 Participants/year) and Injury Risk Ratio of Youth and High School Football Injuries Presenting to US Emergency Departments Stratified by Body Part and Diagnosis

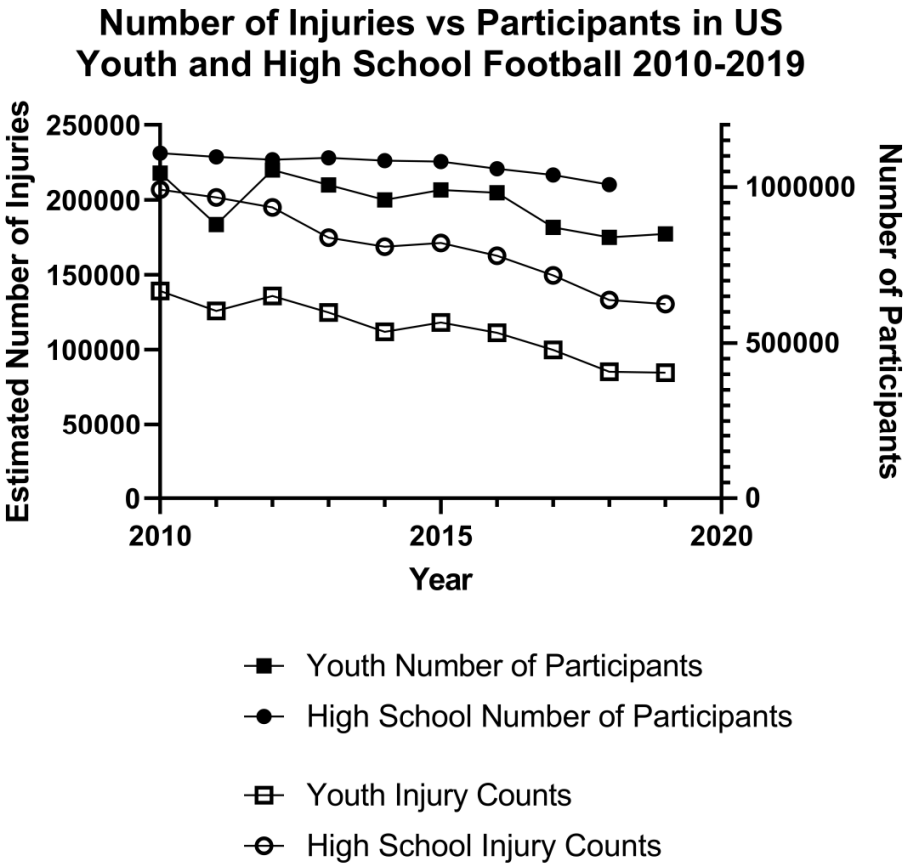
	Youth		High School		
Injury	**Injury Risk**	**95% CI**	**Injury Risk**	**95% CI**	**Injury Risk Ratio**
Body Part					
Head & Neck	2.92	2.49 - 3.30	3.79	3.18 - 4.37	1.30
Shoulder	0.74	0.62 - 0.84	1.74	1.48 - 2.00	2.37
Hand & Wrist	3.31	3.06 - 3.69	3.33	2.89 - 3.76	1.01
Knee	0.93	0.79 - 1.05	1.69	1.44 - 1.93	1.82
Foot & Ankle	1.35	1.15 - 1.53	2.13	1.81 - 2.44	1.57
Diagnosis					
Strain/Sprain	3.09	2.71 - 3.44	4.89	4.25 - 5.50	1.58
Concussion	0.85	0.69 - 1.00	1.68	1.27 - 1.74	1.97
Contusion/Abrasion	2.04	1.77 - 2.27	2.63	2.06 - 2.66	1.29
Dislocation	0.15	0.12 - 0.18	0.78	0.61 - 0.79	5.11
Fracture	2.78	2.38 - 3.13	3.25	2.48 - 3.35	1.17
Internal Injury	0.99	0.77 - 1.19	1.19	0.79 - 1.34	1.20
Laceration	0.49	0.43 - 0.55	0.64	0.50 - 0.65	1.30
All Other Diagnoses	1.58	1.02 - 2.07	2.36	1.40 - 2.82	1.50
Overall	11.97	10.34 - 13.41	16.17	13.85 - 18.38	1.35

ACCEPTED MANUSCRIPT

Table 5. Annual Injury Risk (per 1,000 Participants) and Injury and High School Football Injuries Presenting to US Emergency Departments

	Youth		High School		
Year	**Injury Risk**	**95% CI**	**Injury Risk**	**95% CI**	**Injury Risk Ratio**
2010	13.31	11.23 - 15.38	18.63	16.05 - 21.21	1.40
2011	14.29	11.80 - 16.78	18.36	15.83 - 20.88	1.28
2012	12.87	10.24 - 15.50	17.90	14.64 - 21.17	1.39
2013	12.37	9.68 - 15.07	15.95	12.82 - 19.07	1.29
2014	11.65	8.89 - 14.41	15.53	12.07 - 19.00	1.33
2015	11.90	9.24 - 14.56	15.82	12.31 - 19.32	1.33
2016	11.31	8.47 - 14.16	15.34	11.71 - 18.97	1.36
2017	11.43	8.76 - 14.10	14.38	11.67 - 17.10	1.26
2018	10.14	7.27 - 13.00	13.19	9.94 - 16.44	1.30
2019	9.93	7.24 - 12.62	-	-	-

ACCEPTED MANUSCRIPT



ACCEPTED MANUSCRIPT